STRATA: A Process-Mining Framework for Detecting and Locating HVAC Faults from Building Trend Data

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Abstract—Faults and poor control in heating, ventilation and air-conditioning (HVAC) equipment waste 15 to 30 percent of a commercial building's energy, and many go unnoticed. Building automation systems log the operating data and display it as trend graphs, so a fault is found only when a technician reads the graphs. Automated tools either apply rules written in advance, which miss any fault that was not anticipated, or report anomaly scores that say a day is unusual without saying where or what is wrong. This paper presents STRATA, a framework that works from the trend data a building already records and learns, from one period of normal, fault-free operation, what each day should look like. It then examines every new day with eight checks, each asking whether a physical relationship or an operating sequence has moved outside its normal range. Before any alarm is trusted, every check is tested on fault-free days it did not learn from, to measure how often it raises a false alarm. A check may alarm only when it flags far more days than that rate could explain by chance. Every alarm states what broke, where, the evidence and the check's measured false-alarm rate. The framework is built with process mining: the sensor data is turned into a log of events along the control sequence, models of normal operation are discovered from that log for each terminal unit and for the whole unit, and the per-terminal-unit structure gives every alarm its location. On the five main systems in the public benchmark datasets from Lawrence Berkeley National Laboratory, STRATA detected 158 of 175 seeded faults, most on the first day, with 26 false-alarm days in 480 held-out fault-free days, and named the correct terminal unit in 37 of 50 checkable cases and a wrong one in none. A new building is onboarded by writing one configuration file; the code does not change. A live trial on a building at Thompson Rivers University is the planned next step. STRATA is the first stage of ProcessHeal, a programme whose goal is a building that detects, locates, diagnoses and corrects its own faults.

Index Terms—HVAC, fault detection, process mining, building automation, event logs, explainable alarms

I. Introduction

A building operator today usually learns of a fault in one of two ways: an occupant complains that a room is too hot, or the operator sees that a line on a trend graph has drifted from where it usually sits. Both find the fault late, both depend on someone happening to look, and neither says which piece of equipment is responsible. Buildings account for about a third of energy-related carbon emissions [1], HVAC accounts for up to half of a building's energy use [2], and faults and poor control that nobody sees waste 15 to 30 percent of a commercial building's energy [3]. A study of more than 60,000 pieces of HVAC equipment in commercial buildings found 40 percent of air handlers in a faulted state on any given day, in buildings that already had automation systems with alarms [4].

The tools that exist fall into two groups. Commercial building-automation systems come with libraries of rules written in advance: if the valve is commanded closed but water still flows, raise an alarm [5]. These rules are transparent and need no training data, but they catch only the faults for which a rule was written. Research methods based on machine learning catch more, but they output an anomaly score: a number that says today is unusual without saying where or why [2], which is not enough to write a work order.

This paper presents STRATA, a framework that keeps the transparency of rules and the coverage of learned models. It learns what normal looks like from the building's healthy operation, so it is not limited to pre-written rules. Every alarm it raises states what broke, where, the evidence and the check's measured false-alarm rate, so an operator can act on it. It is built with process mining, a method for discovering a model of a process from a log of events and checking later behaviour against that model [6]. To our knowledge, process mining has not been applied to HVAC fault detection before; the surveys of the field contain no such entry [2], [7].

This report is written so that a reader who has not seen the research paper can understand how the framework works and what it does. Section II describes the framework stage by stage. Section III shows what it did on public benchmark data. Section IV describes how it would be deployed on a real building. Section V states the limitations. The complete evaluation, with every experiment and its outcome, is in the research paper and the public repository.[1]

[1] https://github.com/iyassh/strata

TABLE I
Terms used in this paper.

Term	Meaning
Trend data	The minute-by-minute sensor and actuator values a building-automation system records.
Unit	The equipment scored as a whole: an air handler, a rooftop unit or a fan coil unit.
Zone	A room or group of rooms served by one terminal unit.
Terminal unit	The box that serves one zone: its damper, reheat valve and, in some designs, a small fan.
Event	One thing that happened at one time, such as "cooling started at 09:14".
Event log	All the events of all the days, one case per day.
Process model	A graph, discovered from the log, of the states a terminal unit or a unit passes through on a normal day and in what order.
Conformance	How well one day's events fit the process model.
Healthy band	The range a value or a daily count stayed within on the fault-free training days.
Check	One of the eight tests run on each day; seven can raise an alarm, the eighth only supports one.
Seeded fault	A fault written into a simulation on purpose, so the right answer is known.
Scenario	One simulated run, usually a year, that contains one seeded fault.
Held-out days	Fault-free days kept aside and used only to measure how often each check gives a false alarm; the remaining fault-free days are the training days.
Measured false-alarm rate	The false-alarm rate of a check, measured on the held-out days, and reported with every alarm the check raises.

II. The Framework, Stage by Stage

Figure 1 shows the framework as four tiers. The models and thresholds in the top two tiers are built once per building from one fault-free period; the bottom two tiers, and the event conversion of Stage 2, run on every new day. The stages below follow the figure from top to bottom. Table I defines the terms they use.

A. Stage 1: the building's data and one configuration file

The input is the trend data a building-automation system already records: temperatures, flows, pressures, damper and valve positions, fan speeds, usually once a minute. STRATA does not need new sensors. It needs one fault-free period of that data, ideally a year so that every season is seen, and one configuration file.

The configuration file is the only thing that is specific to a building. It has two parts. The first maps the building's sensor names to the roles the framework understands: for example, which column holds the supply-air temperature and which holds each zone's damper position. The second lists the control sequence for this type of equipment as rules, taken from the design documents, from the ASHRAE Guideline 36 sequences [8] and, as Stage 3 describes, from invariants found in the fault-free data. These rules come in two families. State rules say when the unit is occupied and when it is in heating or cooling mode. Signature rules say which relationships must hold, such as no flow through a closed valve.

Onboarding a new building means writing this file; the framework's code does not change. On the benchmarks, once the framework existed, configuring a second equipment type took about one and a half hours. Two of the five main benchmark systems were configured before any fault data was opened, so nothing in them could have been tuned to the faults.

B. Stage 2: turning sensor data into events

Process mining works on event logs, not on sensor curves, so the first thing the framework does with the data is turn the minute-by-minute values into events along the control sequence. Each day is one case. An event is, for example, "fan started at 06:02" or "cooling became active at 09:14". The configuration's rules produce them: a state rule for operating mode fires an event when the mode changes, and a signature rule fires one when the relationship it guards is broken.

There are two kinds of events, and keeping them apart is a central design decision. State events describe what the equipment is doing, without any judgement of whether it is correct. Signature events describe something that should not happen, such as "water flowing through a valve that is commanded closed". Only the state events are allowed into the next stage, model discovery. If the signature events were allowed in, the discovered model would simply learn the configured fault rules and later "detect" them, which would be circular. We measured the effect: on one system, letting signature events into discovery raised the unit conformance check's apparent detection from 1.8 to 5.2 percent of fault days, and every added day was the model re-finding a rule we had written.

C. Stage 3: discovering what normal looks like

From the state-event log of the fault-free period, the framework discovers process models of a normal day with a standard process-mining algorithm, the inductive miner [9]. It does this at two levels; the levels are the strata the name refers to. The terminal-unit level has one model per terminal unit, describing the normal day of that terminal unit's damper and reheat valve. The unit level has one model for the unit's whole daily cycle (the air handler, in the systems that have terminal units). Each model is a graph of the states the terminal unit or unit passes through and the order it passes through them (a Petri net, in process-mining terms).

This structure does two things in the framework. First, it gives every alarm its location: events, rules and models

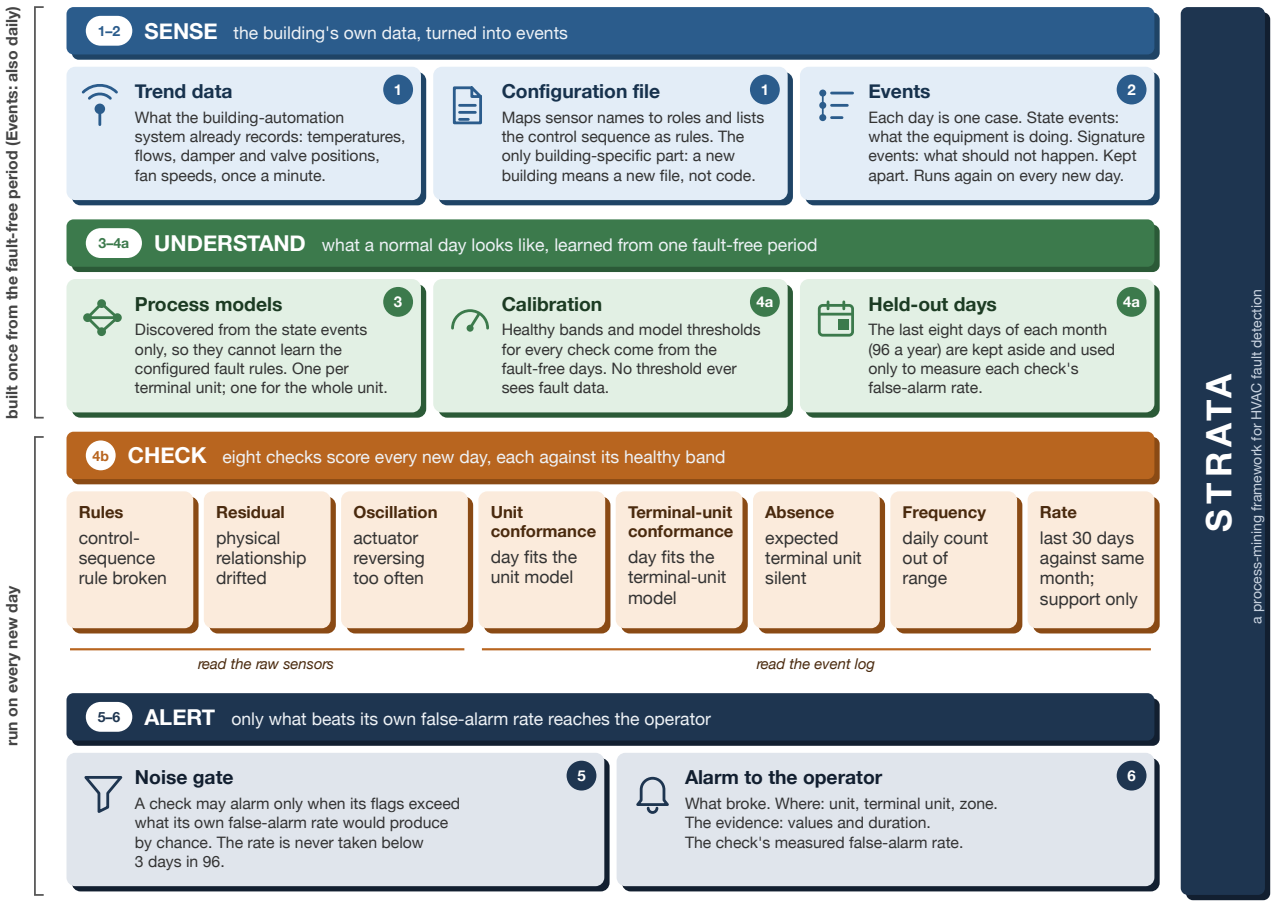


Fig. 1. The STRATA framework as four tiers. The top two tiers are built once per building from one fault-free period: the trend data is turned into events, process models of a normal day are discovered from the state events, and the eight checks are calibrated, with their false-alarm rates measured on held-out days. The bottom two tiers run on every new day, after the day has been turned into events by the same Stage 2 rules: the eight checks score it, the noise gate passes only alarms that beat the check's own false-alarm rate, and what passes reaches the operator stating what broke, where, the evidence and that rate. The numbered badges are the stages of Section II; 4a (calibration) and 4b (the checks) are the two halves of Stage 4.

are all defined per terminal unit, so any check that fires on one terminal unit's data names that terminal unit, and a check that fires on the air handler's data names the air handler. Second, during development the models surfaced invariants of normal operation, things that were always true on fault-free days, which we then wrote down as rules in the configuration. The models do not detect faults by themselves; Section III-D reports the test that established this.

D. Stage 4: calibration, then eight checks

Calibration (4a). Every threshold in every check comes from the fault-free period only. The year is split by calendar date: the last eight days of each month (96 days in a year) are held out and never used for setting anything; they exist only to measure how often each check gives a false alarm. The remaining days, the training days, set the healthy bands and train the models. No threshold ever sees fault data, and no threshold sees the days on which its false-alarm rate is later reported.

The checks (4b). Each day is scored by eight checks, each of which alarms on its own and is named in the alarm it raises. Three of them read the raw sensor data:

- Rules: a control-sequence rule was broken, such as flow through a closed valve, or a zone's airflow far from its setpoint while occupied.
- Residual: a physical relationship that must hold in a known mode has drifted out of its healthy band. With the cooling coil off, the supply-air temperature should equal the mixed-air temperature plus the small rise from the fan; if it does not, a sensor is biased or a coil is leaking.
- Oscillation: an actuator reversed direction far more often than it does on fault-free days, the sign of an unstable control loop.

Five of them read the event log:

- Unit conformance and terminal-unit conformance:

how well today's sequence of events fits the discovered model of the unit or of the terminal unit. A day fits poorly when, for example, cooling starts before the fan, an order the fault-free period never showed.

- Absence: a terminal unit that produces events on every fault-free occupied day produced none on an occupied day.
- Frequency: a daily count, such as the number of heating episodes, left its healthy band.
- Rate: the number of the last 30 days on which a mode was active, compared with the same calendar month of the fault-free year. It never raises an alarm alone, because on one system it added false alarms; it is reported beside another check's alarm as corroborating evidence.

E. Stage 5: the noise gate

A check is allowed to raise an alarm on a scenario only when the number of days it flagged is too large to be explained by its own false-alarm rate. The test is a binomial test, the same calculation used for coin tosses. Suppose a check wrongly flags three of 96 held-out fault-free days. Flagging forty of 365 days in a scenario is then far beyond what chance would give, and the check may alarm; flagging five is within chance, and the check may not alarm. For the test, the rate is never taken below three days in 96, so a check with no false alarms on the held-out days is not treated as perfect. The alarm itself reports the measured count, which can be zero, as in Figure 2.

On the benchmarks the test is applied once per check per scenario, to decide whether the check is credited with the detection; the first day the credited check flagged is the day of the alarm. On a live building there is no scenario, only days. How the test is applied there, for example over a rolling window of recent days, is a decision for the planned trial (step 4 of Table IV). What is already fixed is that every check's false-alarm rate is measured on the held-out days before the first alarm and is reported with each alarm, so the operator knows how often that check raises a false one.

F. Stage 6: the alarm the operator receives

What survives the gate reaches the operator with four things: what broke, where, the evidence, and the check's measured false-alarm rate. Figure 2 shows one, assembled from the framework's recorded output. The fault seeded was a zone damper stuck at 80 percent open. The alarm names the zone, names the terminal unit, states that the primary airflow is 515 cubic feet per minute (CFM) above its setpoint and has been for twelve hours while occupied, reports the damper position of 0.80 against a normal 0.32, and tells the operator that the rules check gave zero false alarms on the 96 held-out fault-free days. It was raised on the first day of the fault.

```
STRATA alarm -- series fan-powered terminal
               units, 2018-01-01 (day 1 of fault)
 what       zone primary-air flow not tracking
          its setpoint            [check: rules]
 where       terminal unit TU_S, zone S
 evidence   715 CFM measured against a 200 CFM
          setpoint, a 515 CFM excess, held for
          720 min while occupied; damper 0.80
          open, 0.32 on the same date of the
          fault-free year
 rate        rules check: 0 false-alarm days on
          the 96 held-out fault-free days
 ---------------------------------------------
 truth       zone-S damper stuck at 80%: location
          and symptom correct, caught on day 1
```

Fig. 2. One alarm from the series fan-powered benchmark. Every value is taken from the framework's recorded output and the benchmark data; the last line is the answer key, not part of the alarm. The alarm names the symptom and the location; it does not claim the root cause.

TABLE II
The seven LBNL datasets. "Scored" is the number of seeded faults evaluated; "Detected" those the framework caught; "False-alarm days" the held-out fault-free days on which it alarmed. †One of these was attributed on audit to a dataset defect rather than to the fault; the totals in the text count it, and are 157 without it. The rooftop datasets are shorter, hence fewer held-out days.

System	Scored	Detected	False-alarm days
Single-duct air handler	14	14†	1 of 96
Fan-powered terminal units (parallel)	30	26	8 of 96
Fan-powered terminal units (series)	29	25	9 of 96
Dual-duct air handler	55	50	3 of 96
Fan coil unit	47	43	5 of 96
Rooftop unit (simulated)	24	20	0 of 28
Rooftop unit (measured)	1 case	—	3 of 49

III. Results on Public Benchmark Data

We evaluated the framework on the public HVAC fault-detection datasets from Lawrence Berkeley National Laboratory [10]. Each simulated dataset is one-minute data for one type of equipment, recorded once fault-free and once per seeded fault: a stuck damper, a biased sensor, a fouled coil, a leaking valve, an unstable controller. The five main systems have a full year of data.

Table II lists the seven datasets: air handlers that serve many zones through one duct (single-duct) or through separate hot and cold ducts (dual-duct), two kinds of fan-powered terminal unit, a fan coil unit that serves one room, and two rooftop units. Six are simulated; one is a measured year from a real rooftop unit. Unless stated, the results below are for the five main systems. The sixth dataset, a simulated rooftop unit, is the only one that records refrigerant-side measurements, so it was used for the faults that are visible only on that side. The measured unit has 182 fault-free days and one fault case, so it yields a false-alarm rate but no detection count.

A. Detection and false alarms

Across the five main systems the framework detected 158 of 175 seeded faults, with 26 false-alarm days in 480 held-out fault-free days, between 1 and 9.4 percent per system. The median time from the start of a fault to its first alarm was one day. On the real rooftop unit the false-alarm rate was 3 of 49 held-out days. A later audit of the configurations mapped every recorded column that the first configurations had left unread; sixteen of the 158 detections came from columns that audit added. The 17 faults it missed are all coil fouling. For some coils every seeded severity was missed, because the recorded sensor values stay inside every healthy band; nothing that reads only those values could see them. On the one dataset that records the refrigerant side, the framework caught every fouling severity.

B. Where the alarm points

On the two fan-powered systems, where each air handler serves four terminal units, 50 detections made before the column audit had their seeded fault in one terminal unit. In 37 of these the alarm named the correct terminal unit, in 13 it named the air handler but no terminal unit, and in none did it name a wrong one. Over all 158 detections the rule carrying the alarm pointed at the right class of component (coil, fan, damper or terminal unit) in 83 percent of cases and at the exact seeded component in 69 percent. The alarm names the symptom and the place. It does not name the root cause: a first diagnosis layer we tested improved localisation slightly but did not reach the targets we had set for it, so we make no diagnosis claim.

C. Against two baselines

Against principal component analysis, a standard statistical method applied to the raw sensors, the framework detected 64 of 73 shared scenarios against the baseline's 61, a difference too small to be statistically significant. The baseline cannot name the terminal unit. Against the Guideline 36 fault rules that commercial systems run, implemented by an independent open-source library and scored under the same gate, the framework detected one more fault than the rules on the single-duct air handler at the same false-alarm count, and three times as many on the terminal units, because it reads the per-zone sensors the rules do not (Table III). We had predicted that STRATA would detect at least as many faults as the rules at a lower false-alarm rate on every system. It did not: on the air handler the false-alarm counts were equal, and on the terminal units the extra coverage cost eight and nine false-alarm days against the rules' none.

TABLE III
STRATA against the Guideline 36 fault rules on the three systems on which both were run, under the same gate. "Detected" is seeded faults detected; "False" is false-alarm days among the held-out days. The single-duct count is the audited 13 of Table II.

System	STRATA Detected	STRATA False	Guideline 36 rules Detected	Guideline 36 rules False
Single-duct air handler	13 of 14	1 of 96	12 of 14	1 of 96
Fan-powered terminal units (parallel)	26 of 30	8 of 96	8 of 30	0 of 96
Fan-powered terminal units (series)	25 of 29	9 of 96	9 of 29	0 of 96

D. What process mining contributes

We began with the hypothesis that the discovered process models would detect the faults themselves. They did not. Over 199 scenarios on six systems (the 175 plus the 24 on the simulated rooftop unit), no fault was caught by the conformance checks alone, and even on the scenario where the models did best, a one-line hand-written rule flagged more fault days than the model did. What process mining contributes is structure. The per-terminal-unit models give every alarm its location. The models surfaced invariants that became detection rules, for example that the air handler's heating never runs on an occupied workday. On the two fan-powered systems the frequency check, which counts events in the log, is one of only two checks (with the oscillation check) that see the unstable-controller faults. Process mining helps to get detection but is not detecting on its own. That finding came from pre-registered tests, with the refuting result written down before each one was run, and it is why detection rests on the calibrated checks while the models supply structure and location.

E. Robustness to noisier data

Simulated data is cleaner than a real building's. To check that this had not inflated the results, we applied four perturbations to the fault-free year and every fault file: sensor noise; change-of-value logging, which records a value only when it changes, as many real systems do; gaps in the record; and shifted schedules. We then re-fitted and re-scored all five systems. Before running this test we had set a limit of 10 false-alarm days in 96. Every system stayed under it, with the worst count at 7 of 96, and each condition cost at most one detection per system.

IV. Deployment on a Real Building: The Next Step

This section describes planned work; none of it has been carried out. Thompson Rivers University (TRU) is the example because the work was done there and a trial is planned. A 2022 energy study of the Campus Activity Centre lists an air-handling unit serving variable-air-volume terminal units, fan coil units and five rooftop units [11]. All are equipment types on which the framework has been evaluated, and nearly all are logged by one Automated Logic building-automation system. Table IV gives the steps.

Three things decide whether it works on a real building, and all three are checked before the first alarm is trusted. First, the fault-free period must really be fault-free: one

TABLE IV
Deploying the framework on a building. Steps 1 to 3 are the top two tiers of Figure 1, set up once; step 4 is the bottom two, with the event conversion of Stage 2, run every day.

Step	What is needed	What comes out
1. Collect	A fault-free period of trend data from the building-automation system at its own sampling rate; a year is best	The fault-free baseline
2. Configure	The point list (every sensor and actuator the system records), mapped to the framework's roles; the control sequence as rules; every recorded point mapped or excluded with a reason	The configuration file, the only file specific to the site
3. Confirm silence	Run the signature rules on the whole fault-free period; any rule that fires there is fixed or removed. The other checks are then calibrated on the training days, and the held-out days give every check's false-alarm rate	A detector whose false-alarm rates are measured on this building's held-out days
4. Operate	Each day's data, turned into events, scored by the eight checks and passed through the noise gate; the window over which the gate counts flagged days is set in the trial	Alarms stating what broke, where, the evidence and the check's measured false-alarm rate

faulty day in it can move a band edge, so the period is audited first. Second, the configuration must read every recorded point; the column audit of Section III showed what unread columns cost. Third, the silence step must run at the building's own sampling rate, because thresholds set on one-minute data do not transfer to fifteen-minute data.

V. Limitations

Every detection result above comes from simulated data, and simulated years are cleaner than real ones; the one real building so far supports a false-alarm rate, not a detection rate. Two scoring choices were made after we saw their effect. The rate check was demoted to support-only, and the two conformance checks, whose thresholds had been set on the same days later used to measure their false-alarm rates, were re-set on separate days, which raised the unit conformance false-alarm rates on the two fan-powered systems. Slowly developing faults are caught late, up to 114 days for two fouling cases. The benchmarks seed equipment and sensor faults, not the schedule and override faults that are most common in real buildings, such as a unit left in manual or a schedule left in holiday mode. Finally, the alarm names the symptom and the location, not the cause; diagnosis and automated correction are the later stages of ProcessHeal, and would need active tests and an operator study before any automated action.

VI. Conclusion

STRATA turns the trend data a building already records into alarms an operator can act on: it learns normal operation from one fault-free period, checks each day against physics and against discovered process models, lets through only alarms that beat a measured false-alarm rate, and states what broke, where, the evidence and that rate. On the five main benchmark systems it detected 158 of 175 seeded faults, with the false-alarm rate of every check measured on held-out days and reported with each alarm, and, in the 50 cases where this could be checked, never named a wrong terminal unit. It is onboarded with a configuration file rather than new code. Process mining gives the framework its structure and the location on every alarm; the detecting is done by the calibrated checks. The next step is a live trial on a real building, and the planned site is the TRU campus.

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